

PhD Researcher in Autonomous Underwater Robotics

Education

PHD IN ELECTRICAL ENGINEERING

National Engineering School of Sfax (ENIS)/ Lab-STA (LR11ES50), Tunisia
Autonomous robotics applied to marine pollution mitigation

2025 - present

- Comprehensive literature review on perception systems for AUVs and ROVs.
- Study of deep learning methods applied to complex underwater environments.

ENGINEERING DEGREE IN ELECTRICAL ENGINEERING (Specialization: Automation and Industrial Computing)

National Engineering School of Sfax (ENIS), Tunisia

2020 - 2023

PREPARATORY CYCLE IN PHYSICS-CHEMISTRY

Higher Institute of Applied Sciences and Technology of Gabes, Tunisia

2018 - 2020

Technical Skills

- **AI and Computer Vision:** OpenCV, Keras, TensorFlow
- **Programming:** Python, C/C++, Java
- **Embedded Systems and IoT:** Raspberry Pi, STM32, ESP32, MQTT communication
- **Tools and Frameworks:** Angular, Spring Boot, MATLAB/Simulink, LabVIEW, Power BI
- **Design:** AutoCAD, Proteus
- **Operating Systems:** Linux, Windows

Professional Experience

PART-TIME LECTURER – Higher Institute of Technological Studies (ISET) | 2025

- Conducted laboratory sessions in **electrical circuits** and **analog electronics**.
- Student supervision and academic assessment.

ELECTRICAL ENGINEER – Construction Work Company | 2024 - 2025

- Performed electrical studies in compliance with standards and technical specifications.
- Designed electrical schematics using **AutoCAD**.
- Supervised on-site electrical installations.

END-OF-STUDIES PROJECT INTERNSHIP – DRÄXLMAIER SATE | 2023

- Developed a full-stack web application to report and track production-line defects.
- Frontend development with **Angular** and backend development with **Spring Boot**.
- Integrated **MySQL** databases and **Power BI** dashboards.
- Conducted API testing with **Postman** and applied **Scrum** agile methodology.

Academic Projects

DRIVER DROWSINESS DETECTION SYSTEM

2023

- Designed a real-time system using **Python** and **OpenCV**.
- Deployed a **Keras** model on Raspberry Pi 4.

EMBEDDED CONTROL AND IOT SYSTEM WITH RASPBERRY PI

2023

- Programmed and controlled sensors and cameras using **Python**.
- Implemented **MQTT** communication and data visualization on **ThingSpeak**.

SMART CONNECTED LOCK USING ESP32

2022

- Developed a smartphone-based access control system (programmed in **C++**).

Languages

- French: Fluent
- English: Upper-intermediate
- Arabic: Native

Additional Training and Activities

- STM32 Training – ACTIA Engineering Services (ARM architecture, UART, Embedded C, STM32CubeIDE).
- Member of the organizing committee for the Electrical Engineering Department Open Day (2023)
- Member of the organizing committee for the ENIS ROBOT 4.0 competition (2022)